

ATISHAY JAIN

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EDUCATION

Brown University

September 2021 - May 2026 (Expected)

Ph.D., Computer Science | Focus: Machine Learning and Foundation Models

GPA: 4/4

IIT (Indian Institute of Technology), Gandhinagar

August 2016 - July 2020

B.Tech, Computer Science and Engineering | President's Gold Medalist

GPA: 10/10

SKILLS

Languages: Python, R, C, Java, C++

Libraries: Pytorch, HuggingFace, Keras, TensorFlow, Sklearn, CUDA, PyTorch Geometric, Spacy, NLTK

Soft Skills: Cross-Functional Collaboration, Project Orchestration, Technical Communication

Generative & Foundational AI: Zero-shot enhancement of Foundation Models, GANs, Auto-encoders, Diffusion frameworks

Model Efficiency and Interpretability: Large-scale inference, memory optimization, Resource-efficiency, Explainers, SHAP, Attention

Sequence & Graph Modeling: NLP, GNNs, LSTMs with Attention, RNNs, Streaming data algorithms (Sketching), Audio processing

GRADUATE RESEARCH PROJECTS

Zero-Shot Enhancement of Foundation Models via Graph Diffusion; [August 2025 - Current]

- Designed a graph diffusion framework to integrate spatial information in spatial transcriptomics data to improve the performance of existing foundation models with no extra training
- Conducted extensive comparative analysis against multiple foundation models and demonstrated superior performance consistently, even in comparison to specialized methods
- [Submitted](#) to RECOMB 2026

Nested Graph Neural Networks for Improved Spatial Clustering; Dr. Ritambhara Singh; [August 2023 - August 2025]

- Implemented a nested Graph Neural Network (GNN), designing a custom architecture for unsupervised clustering of spatial transcriptomics data to enable cell annotation and understand disease pathology
- Validated model performance through rigorous benchmarking against state-of-the-art techniques, GraphST and MuCoST, achieving superior cluster separation
- [Published](#) peer-reviewed research at ACM-BCB 2025 and presented posters at Intelligent Systems for Molecular Biology (ISMB), Montreal 2024 and Cold Spring Harbor Laboratory, 2024 New York, demonstrating strong technical communication and documentation skills

Resource-Efficient Semantic Segmentation via Graph Neural Networks; Dr. Ritambhara Singh; [August 2021 - August 2023]

- Developed a graph neural network-based framework to perform memory-efficient semantic segmentation on large microscopy images
- Reduced computational overhead by 10x–1000x (1–3 orders of magnitude) compared to standard U-Nets, while maintaining competitive pixel-level accuracy benchmarks
- [Published](#) in Microscopy journal and presented at Cold Spring Harbor Laboratory, 2022 New York

Comparative Evaluation Framework for Data-Upscaling Architectures; Dr. Ritambhara Singh; [August 2020 - August 2022]

- Surveyed and reviewed multiple state-of-the-art models that upscale Hi-C data to reduce sequencing costs
- Conducted rigorous performance testing across heterogeneous datasets using multiple quantitative metrics to identify advantages and disadvantages of the models and methods used to train them
- [Published](#) findings in a peer-reviewed publication in the Genes journal

INTERNSHIPS

Caltech (California Institute of Technology); Dr. Mani Chandy; [May 2019 - July 2019]

- Engineered Python algorithms for Caltech's IoTpy framework to process heterogeneous streaming data (sensors, network logs, audio, web-clicks)
- Developed heavy hitter algorithms and an online version of PCA for real-time analysis

IIT Gandhinagar; Dr. Anand Sengupta; [May 2018 - December 2019]

- Built ML de-noising models to identify binary black hole mergers in gravitational wave data

- Analysed neural network models, which use RNNs, LSTMs with attention, Auto-Encoders, and GANs to rapidly de-noise gravitational waves of varying signal-to-noise ratios

Hewlett Packard Enterprise; Mr. Suhas Shivanna; [June 2017 - July 2017]

- Designed a Generic Data Mining Domain Modeller for network telemetry data

SELECTED PROJECTS

Cross-Modal Alignment for Unpaired & Heterogeneous Data; Dr. Randall Balestriero

- Architected a novel multimodal pretraining strategy integrating ViTs and BERT to align heterogeneous unpaired data (images, text, clinical logs) using Gromov-Wasserstein Optimal Transport
- Implemented distributed training pipelines (DDP) and optimized composite loss functions, achieving superior retrieval performance on medical benchmarks with limited supervision compared to CLIP baselines

Defending Neural Networks Against Adversarial Attacks; Dr. Nipun Batra

- Explored and adapted state-of-the-art defenses such as ensembles, defensive distillation, and Defense-GAN against adversarial attacks

Detecting Insults in Social Commentary; Dr. Mayank Singh

- Used text pre-processing (NLP), and ML classifiers such as logistic regression, neural networks and SVMs for offensive comment identification on social media

PUBLICATIONS

- **Jain A.**, Pham, T.M., Laidlaw D.H., Ma Y., and Singh R. Diffusion-based Representation Integration for Foundation Models Improves Spatial Transcriptomics Analysis. Submitted to *Research in Computational Molecular Biology (RECOMB)*, May 2026 [[conference link](#)]
- **Jain A.**, Laidlaw D.H., Ma Y., and Singh R. Improved Spatial Transcriptomics Clustering with Nested Graph Neural Networks. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, October 2025 [[link](#)][[conference link](#)]
- **Jain A.**, Laidlaw D.H., Bajcsy P., and Singh R. Memory-efficient semantic segmentation of large microscopy images using graph-based neural networks. *Microscopy*, Oct 2023 [[link](#)] [[journal link](#)]
- Murtaza G., **Jain A.**, Hughes M., Wagner J., and Singh R. A Comprehensive Evaluation of Generalizability of Deep Learning-Based Hi-C Resolution Improvement Methods. *Genes*, Dec 2023 [[link](#)] [[journal link](#)]
- Gohil V.*, Narayanan S.D.*, and **Jain A***. [Re] One ticket to win them all: generalizing lottery ticket initializations across datasets and optimizers. *ReScience C 6, 2, #4*, Feb 2020 [[link](#)] [[journal link](#)]
- Dutta R.*, Gohil V.*, and **Jain A***. Effect of Feature Hashing on Fair Classification. *ACM India Joint International Conference on Data Science and Management of Data (CoDS-COMAD)*, Hyderabad, India, Jan 2020 [[link](#)] [[conference link](#)]

INVITED TALKS AND POSTERS

Talk at ACM-BCB 2025, Philadelphia - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Lightning Talk and Poster at CSHL Biological Data Science Meeting 2024, New York - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Poster at Intelligent Systems for Molecular Biology 2024, Montreal - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Talk at CSHL Biological Data Science Meeting 2022, New York - Scalable and memory efficient segmentation of large microscopy images using graph-based neural networks

Talk at PyData Gandhinagar 2018 - Applications of Machine Learning Across Domains

HONORS AND RECOGNITION

- Awarded the [President's Gold Medal](#) for obtaining the highest GPA among all B.Tech students graduating in 2020 from IIT Gandhinagar.
- Awarded the [Scholarship for Academic Excellence](#) for obtaining first rank in class, IIT Gandhinagar for the years 2018-19, 2019-20
- Appeared on the [Dean's list](#) (higher than 9.0 Term GPA) for all semesters at IIT Gandhinagar
- Placed in the top 0.2% of IIT-JEE (Joint Entrance Examination), 2016 with over 1.1 million candidates
- Won [HP Code Wars](#), Bangalore, 2015 - a prestigious high-school coding competition conducted by Hewlett Packard where approximately 100 teams participated

SERVICES

- Sub-reviewer for Nature, ICML, NeurIPS, ICLR, and RECOMB conferences.
- Co-founded [Gandhinagar chapter of PyData](#) and organised 4 meetups to expand membership over 1000
- Mentored over 300 students as a teaching assistant across courses related to Python and deep learning.